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Review

Translate or encode? Two design cultures for coupling language and omics data in generative foundation models: a scoping review

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Abstract

Generative foundation models that combine natural language with omics data must decide how biology reaches the language model: by translating it into the model's own tokens or by encoding it with a biological model and injecting the result. In this scoping review (searches to 9 August 2026) we included 55 records describing 51 independent works, 109 models and 585 input routes, decomposed the routes into a library of 22 data operations and charted rationales, supervision, dependence tests and cross-representation comparisons. Seventeen works are token-native, 16 encoder-mediated and 8 hybrid, and 9 use language to condition a biology model. Token-native works keep a median of 100 gene names per cell, and only 2 of the 51 works write measured values as numbers. Works that inject encoder outputs give the language model 1–256 vectors per biological object, and a third of the route steps in the median encoder-mediated work are opaque learned modules. Twenty-five works train or evaluate on text written by another large language model, and only 22 test whether outputs depend on the biological input. In 31 cross-culture comparisons in 20 works, the authors' own representation won 28. Opponents were mostly released, unadapted forms of the other culture, and opponents trained by the proposing team lost all 11 comparisons. The two cultures keep complementary parts of biological data, and the published comparisons cannot yet tell which serves a given task better. We propose a neutral comparison protocol and a reporting card for input representations.

Keywords large language models, foundation models, single-cell transcriptomics, multimodal learning, input representation, scoping review

1. Introduction

Large language models (LLMs) are being extended to biological data: single-cell profiles, nucleotide and protein sequences, histology and microscopy images, molecular graphs and knowledge bases [CITE]. Every such system has to decide how a measurement reaches a network trained on text. Two answers recur (Figure 1a).

- **Translate:** the measurement is written in the language model's own tokens. Examples are a ranked list of gene names, a nucleotide string, or a vocabulary extended with biological symbols. The model needs no new components and can draw on what it learned about named entities during pretraining.
- **Encode:** the measurement is passed through a biological encoder, and the resulting vectors are injected into the language model as soft tokens, through cross-attention or as conditioning. This can keep magnitudes and structure, but the transformation is learned and opaque.

We call these choices design cultures. Each comes with its own conventions for what is kept, what is trained, how outputs are supervised and how claims are tested.

Existing reviews organise text–biology models by architecture family, biological domain or task [CITE]. Individual papers argue for one culture and against the other [CITE]. We are not aware of a cross-paper, evidence-grounded account that asks four questions:

1. what each culture keeps of the data;
2. how each justifies itself;
3. how each is supervised and tested;
4. whether the published comparisons between them can tell which serves a given task better.

This review answers these questions for generative foundation models that couple text with omics data. Text models of protein sequences, protein structures or small molecules are an adjacent family that the search did not target (see Methods and Limitations). To answer the questions, we charted the inputs of every model as grounded routes, decomposed each input path into reusable data operations, and charted design rationales, adaptation, text generated by LLMs, tests of dependence on the biological input and comparisons between representations.

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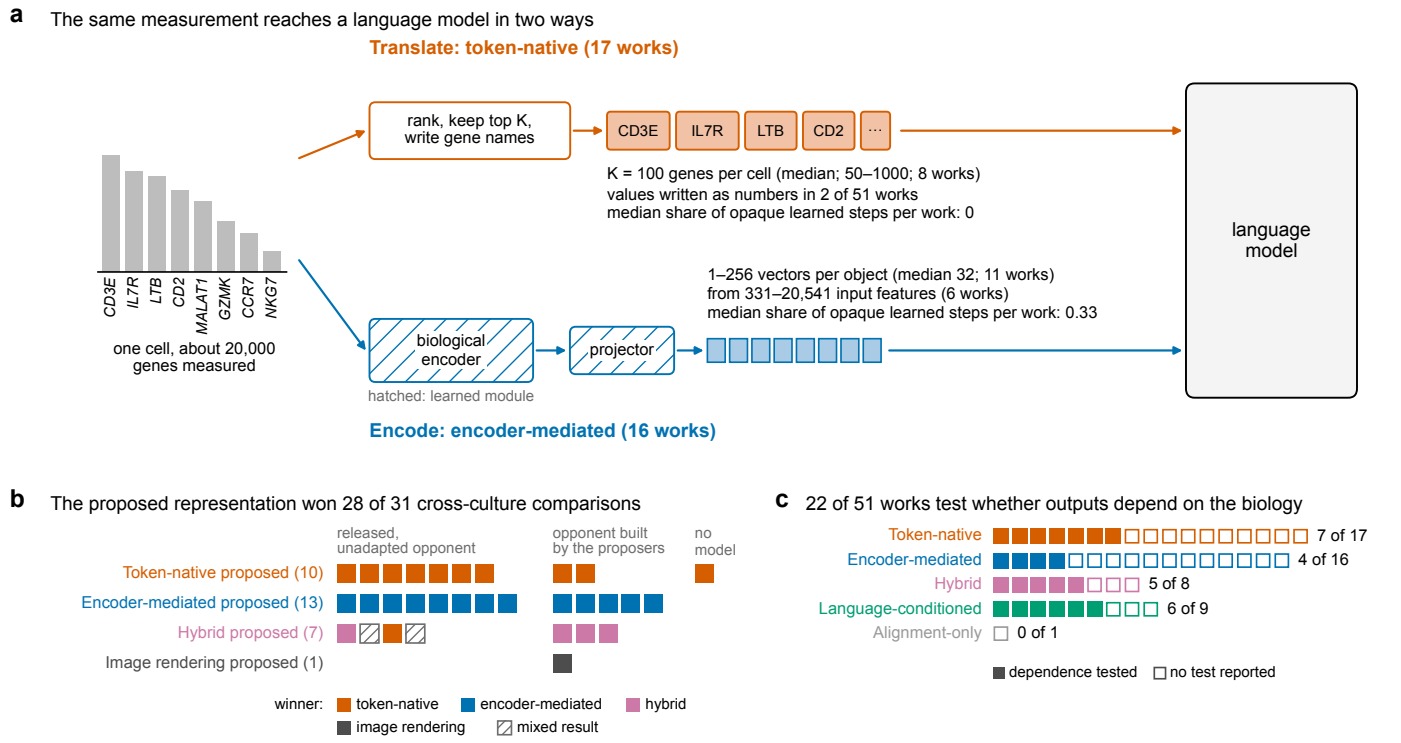


Figure 1 Two ways to give a language model a measurement, and why the published evidence cannot choose between them. (a) The two design cultures, annotated with corpus values. Token-native works rank a cell's genes, keep the top K (the number of genes kept) and write their names into the language model's token stream (K stated in 8 works: 5 token-native, 3 hybrid). Encoder-mediated works pass the measurement through a biological encoder and a projector and inject a few vectors per object (stated in 11 works across cultures; input feature counts in 6). Opaque-step shares are medians over works of the share of biological route steps that are learned modules left unexpanded (14 token-native and 14 encoder-mediated works with decomposed biological routes). Gene names and bar heights are illustrative. Hybrids (8 works) use both lanes; 9 works condition a biology generator on language, and 1 aligns modalities without a generative host. (b) The 31 cross-culture comparison units in 20 works, one square per unit, by the culture of the proposed representation (rows) and how the opponent was obtained (columns). Colour shows the winning culture; hatching marks a mixed result. Released, unadapted opponents are prompted general LLMs and released encoder models used frozen or as published; opponents built by the proposers are baselines or ablations trained on the same data. Image rendering is OpticalDNA's DNA-as-image arm. (c) Works reporting any test of whether outputs depend on the biological input (removal, mismatch or shuffling, random input, a no-biology baseline, content perturbation), one square per work. LLM, large language model.

2. Methods

2.1. Protocol, registration and reporting

The review is reported following the PRISMA extension for scoping reviews (PRISMA-ScR [1]), with PRISMA-S [2] for the search and PRISMA-trAIce [3] for the steps assisted by artificial intelligence (AI). The protocol was not registered before the searches were run. Its first version under version control (version 3.0) was committed to the project repository on 2026-02-09, after a first search execution on 2026-02-06, and the protocol with all later amendments is deposited retrospectively on Zenodo [protocol DOI and deposit date].

The protocol was amended 10 times while the review was carried out, and 12 further changes were made that the protocol did not foresee; Supplementary Table S1 lists each one with its date or stage, its kind and the results that depend on it. The amendments concern the search (top-up rounds, a round that added OpenAlex, abstract enrichment), the screening workflow (role-separated screening, full-text section screening and a human resolution layer) and the procedure for living updates. The further changes added the analyses on which most results rest (the operation-level decomposition, the motif and design-axis analysis, and the design-culture analysis with its targeted charting pass) and record corrections and omissions. Of the further changes, 7 were made after results had been seen: the design-culture analysis with its targeted charting pass, the motif and design-axis definitions, the revision of the decomposition rules after run-to-run disagreement, the replacement of an earlier analysis, the correction that recovered one record, the check of a gap between search windows, and an audit of the full-text screening inputs. Results that depend on the post hoc analyses are described as exploratory where they are reported.

2.2. Eligibility criteria

Included records describe a generative model with foundation-model characteristics that combines a text modality (natural language, or gene tokens processed in a generative language-model style) with at least one biological data modality, published 2018–2026, in English and open access. Records were excluded if they had no biological data modality, no text component or no substantive bridge between text and biology, an encoder-only or otherwise non-generative architecture, or no foundation-model component, and if

they were non-computational, non-scholarly, reviews or duplicates. The full criteria and the screening decision tree are part of the deposited protocol.

Biological data were defined as omics data: single-cell and bulk transcriptomic, chromatin-accessibility, surface-protein, spatial, methylation, epigenomic, genomic and other omics profiles (proteomics, metabolomics). The search vocabulary followed this definition and has no molecule-level terms such as protein, amino acid, DNA or nucleotide. Models whose biological input is a protein or nucleotide sequence were therefore retrieved only when their records also used omics terms.

2.3. Information sources and search

Three concept blocks (biological data AND text or language AND foundation model or generativity) were adapted to each source. The biological-data block lists omics and single-cell terms (for example single-cell, RNA-seq, gene expression, chromatin accessibility, spatial transcriptomics, transcriptomics, genomics, proteomics, epigenomics) and no molecule-level terms. Supplementary Note S1 gives every query string as executed.

Rounds 1–4 searched seven sources: PubMed (NCBI E-utilities), Scopus (Scopus Search application programming interface, API), Semantic Scholar (bulk search API), arXiv (arXiv API; categories cs.LG, cs.AI, cs.CL, q-bio.GN, q-bio.QM and stat.ML), bioRxiv and medRxiv preprints through Europe PMC (REST API), Springer Nature (Meta and Open Access APIs) and Google Scholar (scholarly Python library). Round 5 added OpenAlex and queried Google Scholar with one combined query through a provider export (SerpAPI). The baseline round (window 2018-01-01 to 2026-02-28) was executed on 2026-02-15 under protocol version 3.1; before screening, it replaced an execution of 2026-02-06 under version 3.0. Four top-up rounds covered 2026-03-01 to 04-14, 04-15 to 06-10, 06-11 to 07-06 and 07-07 to 08-09. Each was executed on the last day of its window, except the August round, which was executed on 2026-08-10. Supplementary Table S2 gives, for each round and source, the execution date, the limits as executed and the records returned and retained, and Supplementary Note S2 gives the remaining details of the search.

PubMed, arXiv, Europe PMC, Springer Nature and OpenAlex were queried with exact dates; Scopus and Semantic Scholar were queried by year, after which records outside the window were removed by exact date; Google Scholar was queried by year only. Language and access limits were applied at search in PubMed (English[Language] and "free full text"[sb]), Scopus (LANGUAGE(English) and OPENACCESS(1)) and OpenAlex (filters for English and open access); Semantic Scholar, arXiv, Europe PMC and Google Scholar had no language or access limit. Where no limit was applied at search, English and open access were enforced at screening and full-text retrieval. No published search filters were used. The free Springer Nature API searches full text and cannot be restricted to title and abstract, so records from it were kept only if a term from each concept block appeared in the title or abstract. No study registries, citation searching or contact with authors were used, and the search strategy was not peer reviewed (for example with the PRESS [4] checklist).

2.4. Selection of sources of evidence

Records were deduplicated by conservative exact matching (DOI, PMID, arXiv identifier, normalised title), by matching across rounds against the cumulative record set and by a Crossref audit for hidden duplicates. Missing abstracts were added from Semantic Scholar, Crossref and PubMed; records still lacking an abstract were removed before screening.

Title and abstract screening applied the eligibility decision tree with two role-separated reviewers (scope and architecture) and an adjudicator for conflicts; all three are prompts executed by the same LLM, not independent human reviewers. Full-text screening used machine conversions of the full texts, from which the complete data-source and input-representation sections were reconstructed and screened together with the title and abstract. Records still uncertain after adjudication were resolved by the review lead from the full sections, and the automated decision was kept alongside. The run-to-run stability of this screening is reported in a companion methods paper [CITE companion manuscript].

2.5. Data charting and data items

The unit of charting is a grounded input route: one source object carried through its transformation chain to the representation that the model receives (the carrier). Routes were extracted from the full text with three frozen runs plus dense-coverage candidates and classified by carrier mechanism into five families and 15 subtypes. Biological modality, lifecycle phase, text role and fusion topology were charted as orthogonal dimensions. Dated amendments by the review lead are recorded with the charted data.

Each documented input path was then reconstructed as a directed graph of calls to a library of 22 reusable operations (normalisation, log transform, binning, tokenisation, table lookup, projection, concatenation, attention and others) with named inputs and outputs and instance parameters. Steps whose internal computation the paper does not state remain unexpanded boundaries. A four-paper pilot (ChatNT [5], InstructCell [6], X-Cell [7], MUPAD [8]; 47 trajectories, 509 operation calls) was curated by the authors. The remaining 50 papers were decomposed by an LLM agent in three independent runs per paper, with the structure fixed by the charted routes (one step per element of each route's transformation chain); the published decision for each step is the majority of the three runs, and every decision was checked mechanically for verbatim evidence, valid operation inputs and outputs, and declared parameters. A separate parameter pass then read the whole paper again, with the operations fixed. For every declared parameter of every operation call (2,433 slots) it decided whether the paper states the value for that data path, with verbatim evidence checked mechanically; values that only follow from convention, released code or cited papers do not count. This pass also used three runs per paper and the majority status, and its acceptance gate was fixed before the first run. The decomposition rules were developed on ten length-stratified papers and evaluated on held-out papers with the rules frozen (Supplementary Note S3). The result is version 1.3.0 of the operation library. The withdrawn OKR-Cell [9] preprint was not decomposed.

After a shortlist of candidate findings had been drawn up, a targeted charting pass was run with the review lead's approval. Each of the 54 records was charted in two independent LLM agent runs. Each run received the complete paper text and the list of charted models with their roles, and returned, with verbatim evidence for every item:

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- **representation comparisons:** the arms with their category, the kind and design of the comparison, the task, the metric, the better arm and the stated numbers;
- **tests of whether outputs depend on the biological input:** removal, mismatch or shuffling, random input, a no-biology baseline, or content, order or value perturbation, each with the reported result;
- **evaluation formats** and **outputs**;
- **adaptation:** language-model backbones with size and training state, vocabulary extension, biological encoders and their state, and connectors;
- **LLM-generated text** used for training or evaluation, with the generator, version, prompt and decoding disclosure, and verification;
- **stated rationales** for or against an input representation.

Quotes were checked mechanically against the paper. The variables used for culture-level claims were then adjudicated by reading the evidence: the rationale items, dependence tests found in only one run, and the cross-representation comparison units.

2.6. Critical appraisal

As is usual for scoping reviews, the included works were not appraised for risk of bias, and the certainty of the evidence was not graded.

2.7. Synthesis

We counted models per carrier family, subtype, fusion topology, text role and lifecycle phase (a model is counted once per value it carries), records per publication year, operation reuse as the number of distinct papers per operation, and the share of operation calls whose declared parameters the paper leaves unstated.

Recurring operation sub-chains (motifs) were mined from the decomposed path of every model as closed frequent sub-chains, contiguous or gapped, supported by at least 4 papers, and supported, interpretable sub-chains were named. Each route was also coded on five design axes along one canonical pipeline: carrier formation, interface to the language model, transformation of measured values, text role and transparency. The coding rules are deterministic over the chains and the charted labels, and archetypes are combinations of carrier and interface; a model can carry several. Supplementary Note S4 describes the construction of the paths, the mining, the axes and their tests.

Each included work was assigned to one design culture by how biological data reach the generative model of its proposed system; comparators, data-generation models and components are excluded from this assignment. The cultures are:

- **token-native:** biology reaches a text-pretrained language model only as tokens in its own stream (natural-language names, sequence letters, or new symbolic tokens);
- **encoder-mediated:** biology reaches a text-pretrained language model only as continuous vectors from a biological encoder (soft tokens, attention bridges, conditioning);
- **hybrid:** the same system uses both channels;
- **language-conditioned biology model:** the generative host is a biology model and language enters as conditioning;
- **alignment-only:** no generative host.

The assignment uses the route archetypes and the language-model backbones charted in the targeted pass. Reports of the same study were linked, so all culture-level counts are per independent work (Cell2Text [10, 11], CellWhisperer [12, 13] and OmniNA [14, 15] each have two reports).

A comparison unit is one comparison between arms of different cultures, per work and task family, with the two runs merged. Each unit records:

- its design (a controlled ablation within one system; the authors' system against the same backbone given the data as text; or different systems);
- the arm proposed by the authors;
- the winning culture;
- how the opponent arm was obtained: a prompted general LLM, an external frozen embedding or pretrained model, a trained baseline, a trained ablation, or a representation compared without a model.

For analysis, prompted general LLMs and external embeddings or pretrained models are grouped as released opponents, and trained baselines and ablations as opponents trained by the proposing team. Because units are nested in works, the comparison result is also counted per work (sign test). We used two-sided binomial tests against 0.5 for wins by the proposed arm, two-sided Fisher exact tests for contrasts between cultures and the Mann–Whitney test for per-work shares. The design-culture analysis and the targeted charting pass were defined after the decomposed corpus had been explored, so their contrasts are exploratory and not adjusted for multiple comparisons.

2.8. Use of AI tools

LLMs were run through the Codex command-line agent: gpt-5.4-mini for title and abstract screening, full-text screening and route charting, and gpt-5.6-luna (reasoning effort high) for the operation-level decomposition, the parameter pass and the targeted charting pass, because the earlier model is no longer available through Codex. Full texts were converted with Docling [16] (version 2.110.0) and their sections reconstructed with Docling Graph (version 1.7.0). Prompts, output schemas and rules were frozen and hash-pinned before each production run; every output was validated against its schema, and every evidence quote was checked mechanically against the paper text. Route charting, the decomposition and the parameter pass were each run three times, with the majority decision kept, and the targeted pass twice; agreement between runs was measured, against acceptance thresholds where these had been set. The review lead resolved the full-text records that remained uncertain after automated adjudication and approved the targeted charting pass, and the authors curated the four pilot decompositions. The adjudication of the targeted pass and the culture assignment were made by the library steward, an agent-assisted role in which an AI assistant, directed by the review lead, applied written rules to the verbatim evidence and recorded each decision with its reason; a second, human coder is planned (see Limitations). Supplementary Note S5 gives the tasks, prompts, runs, agreement statistics and human checks in the order of the PRISMA-trAIce items.

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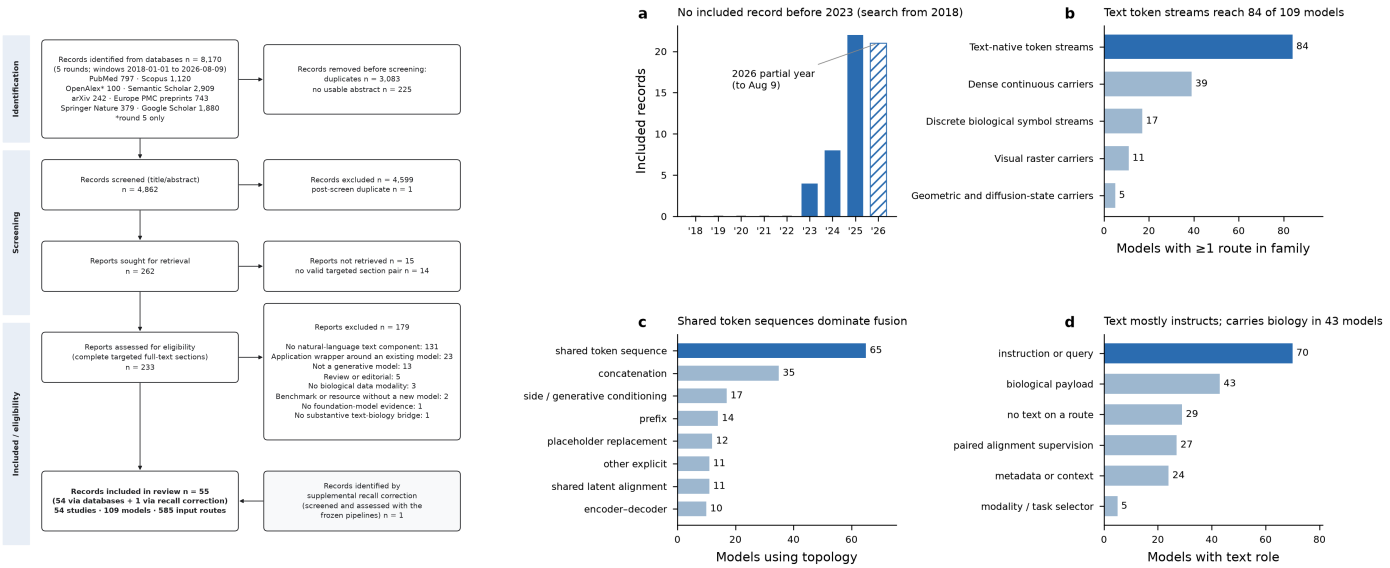


Figure 2 Review flow and corpus. Left: PRISMA 2020 flow with the records identified per database. Right: (a) included records per publication year, (b) models per input-carrier family, (c) models per fusion topology and (d) models per text role. [replace the left panel by a compact flow, keeping the full flow as Supplementary Figure S1, and add the works per culture to panel a.]

3. Results

3.1. Selection of sources of evidence

The searches identified 8,170 records. After removing 3,083 duplicates and 225 records without a usable abstract, 4,862 were screened on title and abstract, and 4,599 were excluded. One post-screen duplicate was removed. Of 262 reports sought, 15 could not be retrieved and 14 lacked a valid targeted section pair, so 233 reports were assessed. 179 were excluded: no natural-language text component (131), application wrapper around an existing model (23), not generative (13), review or editorial (5), no biological modality (3), benchmark or resource without a new model (2), no foundation-model evidence (1), and no substantive text–biology bridge (1) (Supplementary Table S3). Fifty-four records were included from the databases. One more (XunZi [17]) was recovered after an error in the automated keyword check that precedes screening was corrected, and was screened with the same frozen procedures, giving 55 records from 54 studies [5–15, 17–60] (Figure 2, left; Supplementary Figure S1).

3.2. The included works

The 55 records describe 109 models (29 records describe one model; two describe eight), with 467 task/input configurations and 585 grounded routes (Figure 2b–d; Supplementary Table S4). After excluding the withdrawn OKR-Cell preprint, 54 records remain. Three studies have two reports each (Cell2Text, CellWhisperer, OmniNA), so the corpus contains 51 independent works, and the culture-level analyses below count works, not records (Supplementary Table S5).

The records were published in 2023 (4), 2024 (8), 2025 (22) and 2026 to 9 August (21); none is from 2018–2022, although the search started in 2018 (Figure 2a). Years for 13 records added in 2026 top-ups come from the date-bounded search window. The charted lifecycle phases are fine-tuning (53 models), evaluation (51), pretraining (38) and inference (35).

Thirteen included records carry protein inputs (sequences, structures, interaction networks or knowledge bases). Ten of them combine proteins with transcriptomic, nucleotide, methylation or perturbation data. In three, proteins (with small molecules in BioMedGPT [21]) are the only biological input: Chroma [30], PROCYON [50] and BioMedGPT. These three entered through an omics term in the abstract (PROCYON), through Google Scholar (BioMedGPT) or through Scopus, which also searches indexed keywords (Chroma), and are kept as screened.

3.3. How biology reaches the language model: operations, archetypes and cultures

We decomposed 54 papers: 4 pilot papers curated by the authors and 50 papers decomposed by the LLM agent. The decomposition yields 475 input paths, with 1,348 library-operation calls and 819 steps kept as unexpanded boundaries (Figure 3a). 19 of the 22 operations occur in two or more papers. The most widespread are text serialisation (38 papers), selection (26), tokenisation (25), learned projection (23), concatenation (18) and partition (18). In the median paper, 48% of documented steps cannot be decomposed from the text (whole encoders, unstated fusion algebra, training mixtures).

Many choices are not reported (Figure 3b). Of 2,433 parameter slots, 844 (35%) are not stated anywhere in the paper. Among parameters with at least 45 calls, papers left unstated the projection activation in 80% of 76 calls, the concatenation order in 71% of 48, the serialisation order in 42% of 268, the lexical rendering of values in 40% of 268 and the token vocabulary in 26% of 73. Only 52 of the 262 biological routes with library calls (20%) state every declared parameter, and only 4 of 43 works do so for all their biological routes.

Biological routes fall into six main carrier × interface archetypes: encoded soft tokens (24 papers), verbalised text in the token stream (18), generator conditioning (17), biological alphabets in the token stream (8), attention bridges (7) and shared latent alignment (7). The recurring operation sub-chains split along the same line (Figure

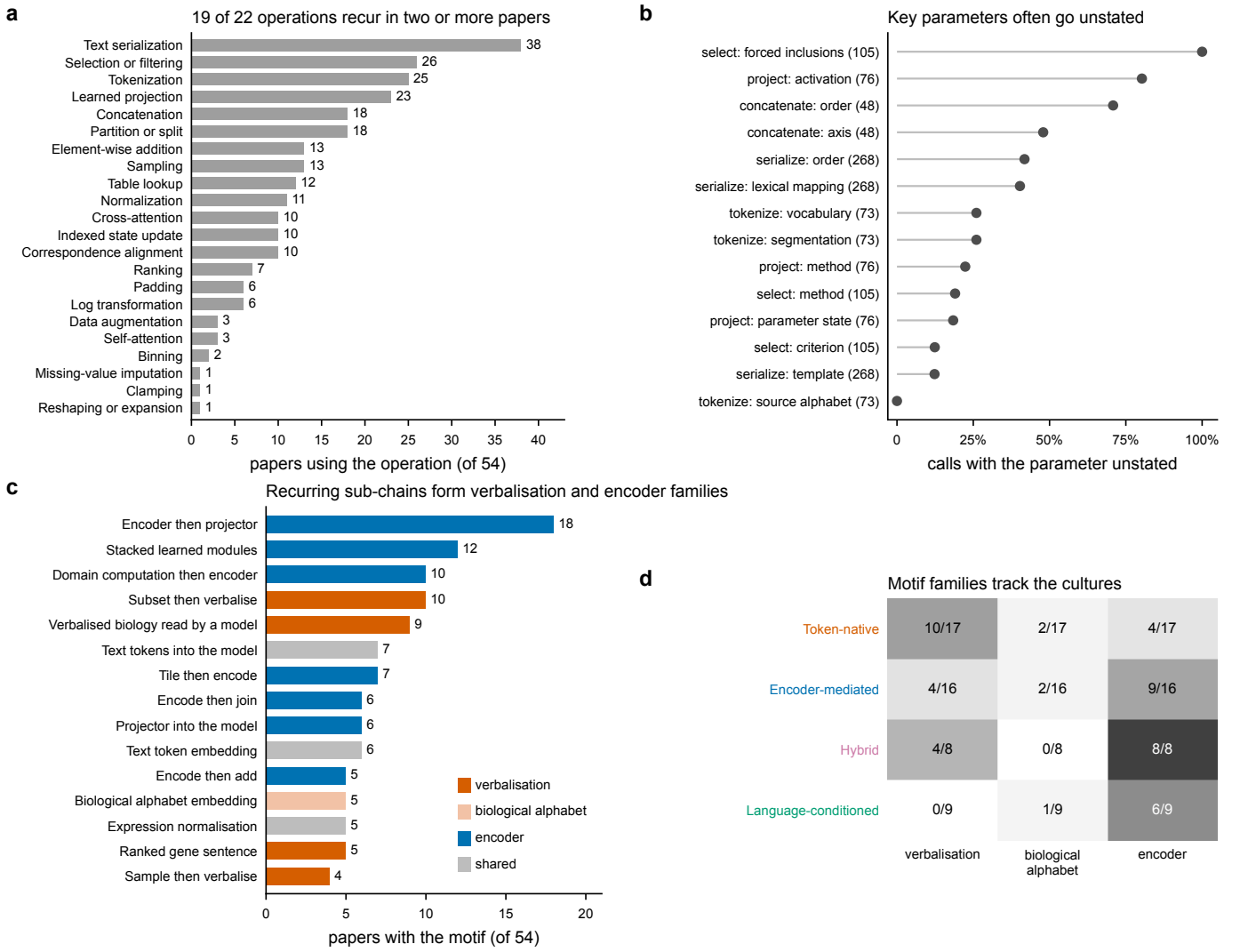


Figure 3 Input paths decompose into reusable operations and recurring motifs. (a) Papers using each of the 22 library operations (54 papers). (b) Share of calls whose parameter is not stated anywhere in the paper, for parameters with at least 45 calls (number of calls in parentheses; agent-decomposed papers). (c) The 15 motifs (closed frequent sub-chains supported by at least 4 papers), coloured by family: verbalisation (subset, rank or sample, then serialise; serialised biology read by a model), biological alphabet embedding, encoder (encoder, projector, stacked or joined learned modules), and shared text-handling or normalisation motifs. (d) Works in which a proposed or component model carries a motif of each family, by culture.

3c; Supplementary Table S6). Verbalisation motifs (subset, rank or sample, then serialise) occur almost only in verbalised inputs. Encoder motifs (encoder then projector, stacked modules, encode-then-join) occur in the encoder-based archetypes. Of models whose motifs are all verbalisation motifs, 16 of 19 have a verbalised primary archetype; of models whose motifs are all encoder motifs, 23 of 28 have an encoder-based one.

At the level of works, this gives two cultures, a hybrid, and a separate family in which language conditions a biology model (Table 1). The assignment of works to cultures was defined after the decomposed corpus had been explored, so the contrasts between cultures reported in this and the following sections are exploratory.

Verbalisation motifs occur in 10 of 17 token-native and 4 of 16 encoder-mediated works, and encoder motifs in 9 of 16 encoder-mediated, 8 of 8 hybrid and 4 of 17 token-native works (Figure 3d). Culture follows the main route by which biology reaches the generative model, so a work can also carry motifs of the other family, for example in auxiliary routes or components.

The cultures are not dictated by the data. Transcriptomics is handled in every culture (token-native 9 works, encoder-mediated 8, hybrid 6, language-conditioned 7), and nucleotide sequences in both main ones (5 and 7). The operations that define each culture are near-universal within it: serialisation occurs in 94% of works with verbalised routes, tokenisation in 75% of works with biological alphabets, cross-attention in 71% of works with attention bridges, and unexpanded learned modules in 57–100% of works in the encoder-based archetypes.

We use the term culture for a bundle of choices that travel together, not for the input route alone. Two parts of the bundle follow from the definition: what reaches the language model and the motif families, which are mined from the same routes. Two do not: the stated justifications, which describe the same trade-off from opposite sides,

Table 1 Design cultures of the 51 independent works.

Culture	Definition	Works
Token-native	biology written into the language model's own stream	17
Encoder-mediated	biology injected as encoder vectors	16
Hybrid	both channels in the same system	8
Language-conditioned	diffusion, flow, graph or sequence generators conditioned on text or text-derived embeddings	9
Alignment-only	no generative host	1

and the choice of opponents, which differs sharply by proposing culture (both reported below). Other practices are shared: rates of LLM-written text, of dependence tests and of fine-tuning by low-rank adaptation (LoRA [61]) or in full are similar in the two main cultures.

Soft tokens are the most or jointly most frequent archetype in every year. The share of papers using verbalisation was 38% in 2024 and 50% in 2026, and biological alphabets rose from 0% to 28%. Yearly counts are small (4, 8, 20 and 18 papers). Hybrids appear from 2024 on (1, 4 and 3 works in 2024–2026).

3.4. What each culture keeps of the measurement

Token-native routes keep identities and order and drop magnitudes (Figure 4a; Supplementary Table S7). Works that write transcriptomes as text keep 50–1000 genes per cell, with a median of 100 over the 8 works that state this number (K). The genes are listed in rank order: Cell2Sentence [38], CellHermes [28], ChatCell [29] and TeamPath [44] keep 100, CellTosg2Sequence [59] 200, and C2S-Scale [24] reports that 1000 often works well. OCellus' [45] spatial prefix keeps 5 marker genes per neighbour. Values are written as numbers in only 2 works (CpG beta values, blood-biomarker values). Rank tie handling is never stated (0 of 24 ranking calls), and selection forced inclusions are stated in none of 105 selection calls. Two works report that within-cell rank tracks log-expression (Cell2Sentence; CellTosg2Sequence, $R^2 \approx 0.90$). If so, what a cell sentence cannot recover is mainly the genes outside the top K (at least 95% of about 20,000 genes when $K \leq 1000$), the order among ties, and any cell-specific departure from the population-level fit between rank and expression.

Encoder routes pass magnitudes to learned modules and hide what survives. Encoder routes in the six works that state their input size use 331–20,541 features (median about 2,800). Eleven works with soft-token or query interfaces deliver each biological object to the downstream model as 1–256 vectors (median 32): 1 (BIOVERSE [23], OmicsLM [46]), 8 (InstructCell, Genolator [36]), 10 discrete codes (RVQ-Alpha [52]), 32 (scMMGPT [41], SciCore-Omics [54], Bio-BLIP [19], scMIR [55]), 200 (CellTosg2Sequence) and 256 (scMOBA [56]). For comparison, Cell2Sentence spends about 350 text tokens on a prompt plus 100 genes. By culture, these counts are: token-native works K = 50–1000 (5 works; hybrids 100–250, 3 works); encoder-mediated works 1–256 vectors (median 32, 7 works) from 2,048–20,541 features (3 works). In the median encoder-mediated work, 0.33 of biological route steps are opaque learned modules, against 0 in the median token-native work (Figure 4b). The projector activation is unstated in 82% of encoder-projector motifs and the concatenation order in 95% of encode-then-join motifs.

Reporting completeness is similar in the two cultures. The median share of unstated parameter slots per work is 0.35 in token-native and 0.26 in encoder-mediated works (Mann–Whitney p = 0.89; Figure 4c). What differs is what is lost (measurement fidelity vs transparency of the transformation), not reporting discipline.

3.5. How each culture justifies its choice

Stated rationales were charted with verbatim evidence and adjudicated. The two cultures describe the same trade-off from opposite sides (Table 2; Figure 5a).

3.6. How the cultures are supervised and tested

LLM-generated text. 25 of 51 works (49%) train or evaluate on text written or rewritten by another LLM (Figure 5b; Supplementary Table S8). In 20 works the generated text carries biological content (descriptions, captions, question–answer pairs, conversations, interpretations, reasoning traces); 4 generate only templates or instructions, and 1 is unclear. Seven works distill reasoning traces from frontier models, 5 report a human check of the generated text, and in 4 works LLM-written text also enters the evaluation. By culture, the counts are token-native 10 of 17, encoder-mediated 8 of 16, hybrid 6 of 8 and language-conditioned 1 of 9. Use of LLM-written text does not differ between the two main cultures (Fisher p = 0.73) and is rare only where language conditions a biology generator (1 of 9 vs 24 of 42 other works, p = 0.024). It is a property of language-model-hosted designs, not of either culture.

Adaptation (works where both runs agree). Token-native works fine-tune the language model fully (9) or with LoRA (6), train it from scratch (3), and extend its vocabulary (4). Only encoder-mediated works keep the language model frozen (3 of 16, against 0 of 17 token-native works; Fisher p = 0.10). They freeze (6) or fine-tune (6) the biological encoder.

Is the biology used? Twenty-two of 51 works (43%) report any test of whether outputs depend on the biological input: removal, mismatch or shuffling, random input, a no-biology baseline, or content perturbation (Table 3; Figure 1c).

The rates do not differ detectably between cultures (encoder-mediated vs all other works, Fisher p = 0.13; token-native vs encoder-mediated, p = 0.46); the low overall rate is the finding. Bio-BLIP's authors argue that linear projection interfaces encourage reliance on prompt text, which would make such tests most informative for encoder-mediated designs. Works that made cross-culture comparisons tested dependence no more often than the others (10 of 20 vs 12 of 31, Fisher p = 0.56).

Evaluation formats (works, both runs / either run): closed-set classification 31 / 42; open-text generation scored by automatic text metrics 29 / 43; likelihood or perplexity scoring 9 / 13; candidate-label selection 6 / 11; open text judged by an LLM 6 / 7, or by humans 6 / 8; wet-lab validation in 1 work.

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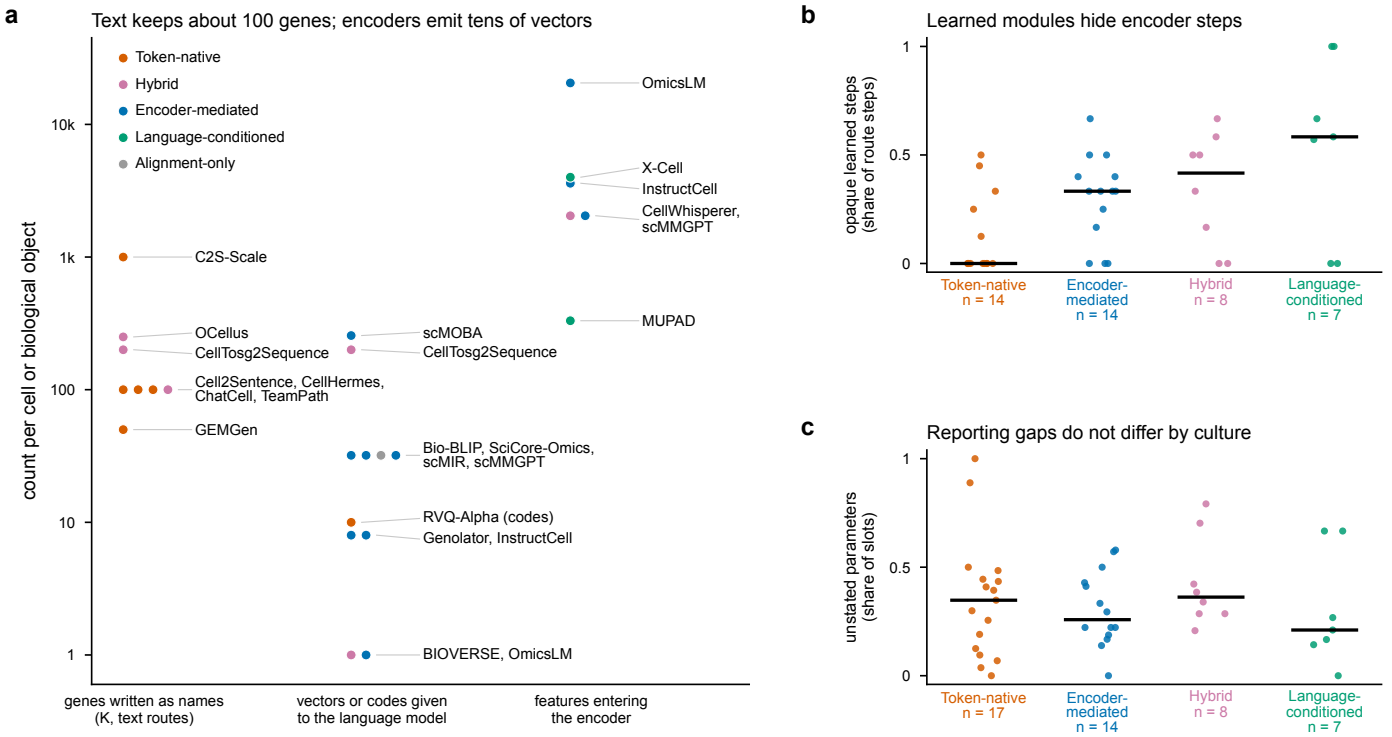


Figure 4 What each culture keeps of a measurement. (a) Counts per cell or biological object as stated in the papers: genes kept when transcriptomes are written as names (proposed-model text routes; 8 works), vectors or discrete codes passed to the language model (11 works; RVQ-Alpha uses learned discrete codes in the token stream), and features entering the encoder (6 works). Points are works coloured by culture; works with the same value share a label. For comparison, Cell2Sentence reports about 350 text tokens for a prompt plus 100 genes. (b) Per-work median share of biological route steps that are opaque learned modules (works with agent-decomposed biological routes). (c) Per-work share of parameter slots unstated in the paper (proposed and component models; token-native vs encoder-mediated, Mann–Whitney $p = 0.89$). Bars are medians; the alignment-only work is omitted from b and c.

Table 2 Stated rationales for and against input representations, by culture (works; adjudicated from verbatim quotes).

Rationale	Token-native (of 17)	Encoder-mediated (of 16)
Unchanged architecture	9 (e.g. C2S-Scale "avoids modifying the underlying architecture of LLMs")	none in both runs
Language model's prior knowledge of named entities	9	—
Concedes loss of quantitative or fine-grained information in its own representation	7 (e.g. Cell2Sentence, CellHermes, ChatCell, RVQ-Alpha)	—
Criticises text rendering for losing magnitudes, low-abundance genes or structure	—	7 (Cell2Text, BIOREASON [22], InstructCell, scMMGPT, scMOBA, Bio-BLIP, OmicsLM)

Table 3 Works reporting a test of whether outputs depend on the biological input, by culture.

Culture	Works with a dependence test
Token-native	7 of 17
Encoder-mediated	4 of 16
Hybrid	5 of 8
Language-conditioned	6 of 9

3.7. What the head-to-head comparisons show

The charting runs returned 217 comparison items. Adjudication merged them into 31 comparison units in 20 works whose arms belong to different cultures (Supplementary Table S9). Three further units were set aside as within-culture or not about representation form.

Home advantage (Figures 1b and 5c–d). The authors' own representation won 28 of 31 units: 20 of 23 between-system comparisons (87%; two-sided binomial $p = 4.9 \times 10^{-4}$ against 0.5), all 7 controlled ablations and the one same-backbone comparison. Units are nested in works (C2S-Scale contributes four). Counted by

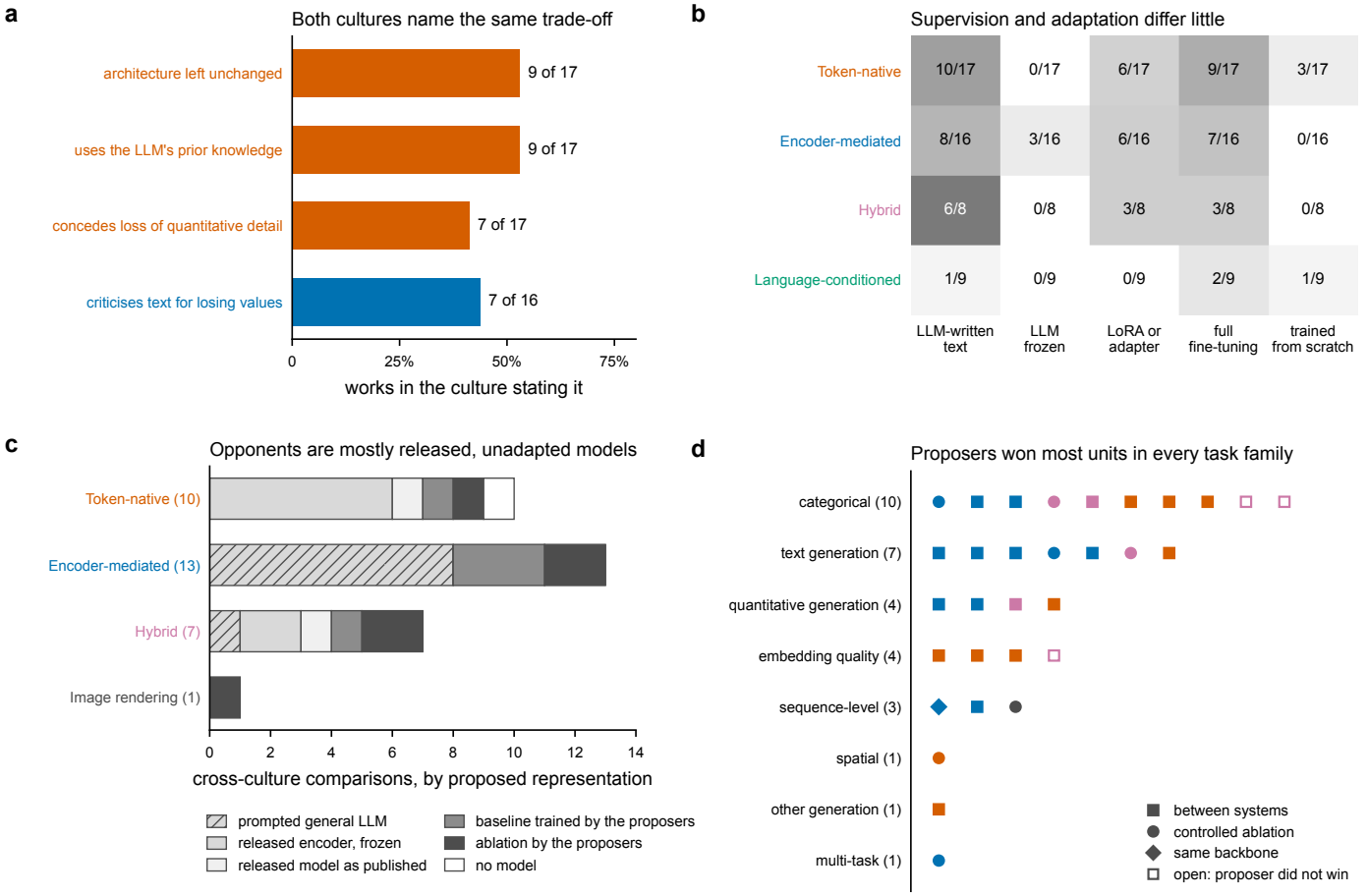


Figure 5 How the cultures justify and test their choice. (a) Stated rationales, adjudicated from verbatim quotes: for the work's own representation (token-native works) or against text rendering (encoder-mediated works). (b) LLM-written text and adaptation of the language model by culture (cells: works / works in the culture; adaptation counted where both charting runs agree). (c) Opponent type in the 31 cross-culture comparison units, by proposed representation. (d) Units by task family; marker shape shows the design and colour the proposed culture; open markers are the three units the proposer did not win (two mixed, one loss). Contrasts between cultures are exploratory. LLM, large language model.

work, the proposer won every unit in 17 of 20 works (sign test $p = 0.0026$) and every between-system unit in 14 of 17 ($p = 0.013$).

Across papers the winners split between cultures (between-system winners: token-native 10, encoder-mediated 9, hybrid 2, mixed 2), so no culture dominates. Within every task family the winner followed the proposer; the non-wins fell in categorical tasks (2 of 10 units) and embedding quality (1 of 4), and none in text generation (7 units), quantitative generation (4) or sequence-level tasks (3).

Each culture tests mostly against released, unadapted forms of the other.

- In encoder papers, the opponent was a prompted general LLM reading raw text in 8 of 13 units: GPT-4.1, GPT-4o [62] or Llama-2-Chat [63] given sequences, SMILES strings or gene lists without adaptation.
- In token-native papers, it was a released encoder-based model in 7 of 10 units, usually a frozen embedding with a simple head (scGPT [64], Geneformer [65], ESM-2 [66], Nicheformer [67]) or a model used as published (CellWhisperer).
- The opponent type differs sharply by proposing culture (prompted LLMs 8 of 13 vs 0 of 10, Fisher $p = 0.003$; released encoder models 0 of 13 vs 7 of 10, $p = 4.9 \times 10^{-4}$). This is partly by construction, since a cross-culture opponent must come from the other culture; the choice of an unadapted opponent is not.
- Opponents trained by the proposing team on the same data (baselines or ablations) made up 5 of 13 units in encoder papers and 2 of 10 in token-native papers. Across all cultures the proposer won all 11 such units and 16 of 19 units against released opponents; the remaining unit compared representations without a model. All three exceptions involved released opponents.

What the comparisons show. When the token-native arm read raw sequence letters, the encoder arm won 5 of 7 units; the exceptions were a rendering of DNA as images (OpticalDNA [49]) and a vocabulary trained on biological tokens (OmniGene-4 [48]). For gene-name lists, the winner followed the proposing culture (token-native 8, encoder-mediated 7, hybrid 3, mixed 2). The three exceptions were:

- a 120B general model reading 128 gene names beat the 8B hybrid BIOVERSE on zero-shot annotation (accuracy 0.779 vs 0.614), while an 8B general model did not (0.369);
- C2S-Scale-27B beat the 32B hybrid CellTosg2Sequence on immune annotation (exact 85.6% vs 24.1%); the CellTosg2Sequence authors attribute the gap to C2S-Scale's much larger pretraining corpus, against about 8,000 annotation examples in their own reinforcement stage;
- OCellus embeddings won a linear probe but not clustering.

In controlled ablations, where only the representation changed, the encoder or hybrid arm won whenever the authors had proposed it: expression values vs cell sentences in scMMGPT (annotation error 0.76% vs 6.42%), aligned features vs a raw sequence in the same Llama-2-Chat (BioMedGPT; BLEU-2 0.344 vs 0.019), and an embedding stack vs a Cell2Sentence-like control (OmicsLM; average relative rank .740 vs .090). The token-native arm won the one controlled test its authors proposed: numeric coordinate tokens vs learned positional embeddings in TISSUENARRATOR [58], reported qualitatively.

Hybrids. Eight works combine both channels, and 11 deliver one modality through two carrier formations. In both controlled tests the hybrid beat its single channel: CellWhisperer (an embedding plus the top-50 gene list, a mild benefit) and CellTosg2Sequence (knowledge-graph tokens plus a cell sentence, 0.385 vs 0.268 fuzzy accuracy). In both cases the hybrid was the proposed configuration. A related within-culture result points the same way: in RVQ-Alpha, adding gene evidence in text to learned discrete codes gained 8.3 percentage points and reduced hallucinations.

4. Discussion

4.1. Principal findings

Generative text–omics foundation models are built in two design cultures:

- **translating** biology into the language model's own tokens;
- **encoding** it with biological models and injecting vectors.

A growing hybrid combines both, and a separate family uses language to condition biology models. The cultures are not imposed by the data: transcriptomes and sequences are handled both ways.

The cultures keep complementary parts of a measurement:

- Token-native inputs keep identities and order, and lose explicit values and every gene outside the top K.
- Encoder-mediated inputs pass values and structure to learned modules, so what survives is not visible and is rarely tested.

Where authors discuss the choice, they describe this trade-off the same way from both sides: token-native works cite unchanged architectures and the language model's knowledge of named entities and concede the loss of quantitative information, while encoder-mediated works criticise text rendering for that loss. The distinction itself is familiar. What has been missing is a measurement of its consequences and evidence about when each side of the trade-off matters.

4.2. Why the published comparisons cannot settle the question

Three features of the evidence keep it from discriminating between cultures.

1. **Home advantage.** The proposing culture won 28 of 31 cross-culture comparisons (every comparison in 17 of 20 works), including every controlled ablation. Selective publication alone would produce this pattern, so these wins cannot rank the cultures.
2. **Unadapted opponents.** Each culture tests mostly against released, unadapted forms of the other:
   - encoder papers against general LLMs that read raw sequences or gene lists without adaptation;
   - token-native papers against released encoder models, usually frozen embeddings with simple heads.

   When both arms are trained on the same data, the comparison is an ablation or baseline built by the proposers, who also decide what to ablate; the proposer won all 11 such units.
3. **Rare dependence tests.** Only 22 of 51 works test whether outputs depend on the biological input, with no detectable difference between cultures. Without such tests, a win cannot be attributed to the representation rather than to prompt text, metadata or the language model's prior knowledge.

The few results that went against the proposer are informative. They suggest that verbalised gene lists can be competitive for categorical identity tasks such as cell annotation when the reading model is large or trained on a great deal of single-cell data. Encoders won most units (5 of 7) in which the text arm had to read raw sequence letters. These are hypotheses for a controlled study, not conclusions.

4.3. A neutral comparison protocol

We propose that claims about input representation be supported by comparisons that fix everything except the representation (Box 1).

4.4. A reporting card for input representations

Reporting completeness is similar across cultures, but the missing items differ (Box 1).

Box 1 A neutral comparison protocol and a reporting card for input representations

Neutral comparison protocol (everything fixed except the representation)

- **Matched systems:** the same backbone, data, training budget and adaptation (LoRA or full fine-tuning) for every arm.
- **Matched information budget:** report the features kept and the tokens per object for each arm, and vary K for verbalised arms.
- **Tasks at both levels:** identity-level tasks (annotation, retrieval) and magnitude-level tasks (perturbation response, expression or dose prediction, variant effect).
- **Dependence controls:** removal, mismatch and shuffling of the biological input, plus a no-biology baseline with the same backbone; for verbalised inputs, also gene-name scrambling.

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- **Like-for-like baselines:** when a published method is used as a comparator, use its documented settings (for example the number of genes in a cell sentence), or report both.

Reporting card. Items to report for each input representation:

Culture	Card items
Token-native	K, selection criterion, ordering and tie rule, normalisation, template, lexical rendering of values, vocabulary or segmentation
Encoder-mediated	encoder and its training state, projector method and activation, number of vectors per object, which components are trained, concatenation order and axis
LLM-generated supervision (all cultures)	generator and version, prompts, decoding parameters, verification

The operation library provides a controlled vocabulary for these items.

4.5. Relation to existing reviews

Recent surveys classify models by family, modality and task; LLM4Cell [68], for example, groups 58 single-cell methods into five methodological families and maps them to eight analytical tasks. [CITE, to verify: ACL 2025 survey of single-cell foundation language models; LLMs meet virtual cell; Nature 2025 perspective on multimodal foundation models in molecular cell biology; 2026 Cell Systems compositional-FM review; scoping review of LLM-generated synthetic data in biomedicine.] The translate-or-encode distinction is familiar to primary authors, who use it to justify their designs. This review measures, at the level of operations, what each design keeps and how its claims are tested, across omics modalities.

4.6. Implications

For builders, the current evidence supports working hypotheses rather than rules:

- identity-level tasks may be served by verbalised inputs read by strong language models;
- tasks that depend on magnitudes or raw sequence may favour encoders;
- hybrids beat their single channels in both controlled tests, both run by their proposers.

Each hypothesis can be tested with the protocol in Box 1.

For reviewers and benchmark designers, the priority is neutral, budget-matched comparisons with dependence controls, rather than further between-system leaderboards.

4.7. Limitations

Automation and human checks. Screening, charting and decomposition were assisted by LLMs, and the role-separated screening prompts are not independent reviewers. The decomposition graphs are three-run consensus decisions, not author-verified reconstructions, and were not checked by a reviewer agent. Decomposition stability is incomplete. On unseen papers, the step-level consensus was reproduced in 95.6% of steps but complete route sequences only in 85.4%, below the predeclared 90% gate. Among the three production runs on all 50 agent-decomposed papers, five papers had a pairwise operation-set Jaccard below the predeclared minimum of 0.75 (one at 0.0), so the full-corpus gate was not met either. Residual disagreement concerns how prompts are merged with metadata (serialisation vs concatenation) and whether a named "tokenized" representation counts as a tokenisation step. The rules were tuned on development and earlier held-out papers, so in-sample stability is higher than out-of-sample stability. Steps are bounded by the charted routes, so errors in charting carry over. Parameter status was decided by agents, not authors. Held-out consensus agreement was high (0.992), but single-run agreement on the 30 papers outside the development and held-out sets was lower (0.912 vs 0.956 held-out), and one run needed a clean restart after a quote failed validation. "Unstated" means not found in the paper text; values in code or cited works are not credited.

Search and selection. Open access and English were required. At search they were applied only in PubMed (with "free full text" as a proxy for open access), Scopus and OpenAlex, and elsewhere at screening. Fifteen reports could not be retrieved and 14 had no valid targeted section pair. Google Scholar was incomplete in three rounds (5 of 7 queries in April and June, no records in July), and one eligible record was found only after a recall correction. The executed search windows have a 13-day gap: the baseline was run on 2026-02-15 with a window to 2026-02-28, and the first top-up started at 2026-03-01, so the date-bounded sources were never queried for records dated 2026-02-16 to 2026-02-28. Google Scholar's year-level queries in later rounds and dates that differ between databases reached part of this window. On 2026-10-14 we re-ran the baseline queries of PubMed, arXiv, Europe PMC and Semantic Scholar for the window; they returned 51 unique records, 17 had already been retrieved, and none of the other 34 met the eligibility criteria in an indicative single-pass LLM title/abstract triage. Scopus and Springer Nature were not re-run. In the baseline full-text screening, the abstract shown with the selected sections was cut to about 1,500 characters for 73 of 221 records (selected sections were complete), and five reports whose only reconstructed section covered at least 90% of the paper were not screened at full text; the author judged the risk of missed inclusions low and kept the included set. The withdrawn OKR-Cell preprint is still counted in the 55 records (author decision pending). The search closed on 2026-08-09, and no refresh was done for this manuscript.

Scope. The search vocabulary covered omics data only. A coverage check found that 12 protein and biomolecular text–generative models (ProteinDT [69], BioT5 [70], BioT5+ [71], NatureLM [72], Evolla [73], Galactica [74], InstructProtein [75], ProtLLM [76], Prot2Text [77], ProtT3 [78], ProLLaMA [79], TxGemma [80]) were never retrieved; none has a term of the biological-data block in its title or abstract, and protein or nucleotide blocks would have added many records (Supplementary Note S6). The findings therefore describe omics-framed models. Models whose only biological input is a protein are represented by three works, and nucleotide-sequence models described without omics terms are under-represented. Whether the two cultures and the home advantage hold for these families is untested.

Analysis. Motifs are mined from route chains bounded by the charted transformation chains and from four author-curated pilot graphs, so steps that the charting did not record cannot form motifs. Support thresholds (at least 4 papers) and motif naming involved author judgement. The design-pattern axes are derived from the charted routes and their decomposition. Interface and carrier

labels inherit charting errors, and the axis for the transformation of measured values falls back on the charted carrier subtype for 90 biological routes whose chain has no decided operation. The axes were defined by the authors after inspecting the data and were not validated on an independent corpus. Protocol objective 3 (datasets, downstream tasks, evaluation metrics) and the extraction fields for pretraining objective and code/data availability are not yet charted, except the evaluation formats and outputs recorded in the targeted pass. The design-culture analysis was defined after exploring the decomposed corpus, so its hypotheses are post hoc. Its charting used two runs of one model family and adjudication by a single steward. Run agreement was high for comparison presence (κ 0.92), LLM-generated text (κ 0.96) and dependence tests (Jaccard 0.91), but lower for evaluation formats (0.57) and rationales (0.60). A second human coder for the 31 comparison units and the culture assignment is planned before submission. Culture assignment for works that combine separate models is a judgement call; Geneverse [33] is coded token-native, although its image models are encoder-mediated. Group sizes are small (17, 16, 8 and 9 works), and culture contrasts are exploratory (unadjusted Fisher tests). Comparisons are as reported by the authors; publication bias contributes to home advantage, so it is part of the finding as well as a limitation. Comparisons reported only in figures can be missed by the charting runs. While checking one exception (CellTosg2Sequence), we found a Tahoe-100M [81] perturbation-classification comparison against C2S-Scale-27B, won by the proposer, that neither run charted; it is not counted. Such misses would add to home advantage, and the second-coder check will include a recall audit of comparisons. Gene and token budgets are taken from stated values in 8 and 11 works; works that do not state them are not counted. Earlier keyword pre-screens of the full texts (used to choose the variables for the targeted pass) skipped appendices after the reference list. All reported counts come from the targeted pass, which used the complete text.

5. Conclusions

Generative text–omics foundation models are built in two design cultures that keep complementary parts of a measurement: token-native inputs keep identities and order, and encoder-mediated inputs pass values and structure to learned modules whose transformation is not visible. Each culture is tested mostly against released, unadapted forms of the other, and few works test whether outputs depend on the biological input, so the published comparisons cannot yet tell which serves a given task better. Neutral, budget-matched comparisons with dependence controls, and complete reporting of input representations, are needed before either culture can be preferred for a task.

Key Points

- Generative text–omics foundation models follow two design cultures: token-native models write biology into the language model's own tokens, and encoder-mediated models inject vectors from biological encoders; hybrids combine both.
- Token-native inputs keep gene identities and their order (a median of 100 genes per cell) and rarely keep measured values; encoder-mediated inputs pass values to learned modules whose transformation is not visible.
- Only 22 of 51 works test whether outputs depend on the biological input, and 25 train or evaluate on text written by another large language model.
- The proposing team's representation won 28 of 31 cross-culture comparisons, mostly against released, unadapted opponents, so the published evidence cannot rank the two cultures.
- A neutral comparison protocol (matched systems and information budgets, identity- and magnitude-level tasks, dependence controls) and a reporting card for input representations are proposed (Box 1).

Supplementary data

The Supplementary Information contains the supplementary notes, tables and figure cited in the text, and the data files with a column dictionary.

Data and code availability

The search configurations and exports, screening and charting outputs, the operation library (version 1.3.0), the analysis code and the source data of every figure panel are available at https://github.com/BogdanDidenko/text-bio-fundational-models-review (this version: commit 5f27f9fb). `manuscript/supplementary/README.md` gives the repository path of every supplementary item, and the earlier version of this manuscript, with its internal labels, is `manuscript/archive/review_manuscript_draft_v0.5.1.md` (commit 4c4db9f0). The component library is browsable at https://bogdandidenko.github.io/text-bio-fundational-models-review/component-library/. A curated snapshot of the repository and the protocol with its amendments are deposited on Zenodo [DOIs].

Use of large language models

Screening, charting and the operation-level decomposition used LLM agents, as described in Methods (Use of AI tools) and Supplementary Note S5. [The authors state here any use of language models in drafting or editing this manuscript, as required by the journal's policy.]

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Author contributions

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References

1. Tricco AC, Lillie E, Zarin W, O'Brien KK, Colquhoun H, Levac D, et al. PRISMA Extension for Scoping Reviews (PRISMA-ScR): Checklist and Explanation. Annals of Internal Medicine. 2018;169(7):467–473. doi:10.7326/m18-0850.
2. Rethlefsen ML, Kirtley S, Waffenschmidt S, Ayala AP, Moher D, Page MJ, et al. PRISMA-S: an extension to the PRISMA Statement for Reporting Literature Searches in Systematic Reviews. Systematic Reviews. 2021;10(1). doi:10.1186/s13643-020-01542-z.
3. Holst D, Moenck K, Koch J, Schmedemann O, Schüppstuhl T. Transparent Reporting of AI in Systematic Literature Reviews: Development of the PRISMA-trAIce Checklist. JMIR AI. 2025;4:e80247–e80247. doi:10.2196/80247.
4. McGowan J, Sampson M, Salzwedel DM, Cogo E, Foerster V, Lefebvre C. PRESS Peer Review of Electronic Search Strategies: 2015 Guideline Statement. Journal of Clinical Epidemiology. 2016;75:40–46. doi:10.1016/j.jclinepi.2016.01.021.
5. de Almeida BP, Richard G, Dalla-Torre H, Blum C, Hexemer L, Pandey P, et al. A multimodal conversational agent for DNA, RNA and protein tasks. Nature Machine Intelligence. 2025;7(6):928–941. doi:10.1038/s42256-025-01047-1.
6. Fang Y, Deng X, Liu K, Zhang N, Qian J, Yang P, et al. A Multi-Modal AI Copilot for Single-Cell Analysis with Instruction Following; 2025. arXiv:2501.08187.
7. Wang C, Karimzadeh M, Ravindra NG, Bounds LR, Alerasool N, Huang AC, et al. X-Cell: Scaling Causal Perturbation Prediction Across Diverse Cellular Contexts via Diffusion Language Models. bioRxiv; 2026. Preprint. doi:10.64898/2026.03.18.712807.
8. Xiang J, Li M, Hou S, Chen Y, Luo X, Ji Y, et al. A Generative Foundation Model for Multimodal Histopathology; 2026. arXiv:2604.03635.
9. Wang H, Zhang X, Fang S, Ran L, Deng Z, Zhang Y, et al. WITHDRAWN: OKR-Cell: Open World Knowledge Aided Single-Cell Foundation Model with Robust Cross-Modal Cell-Language Pre-training. bioRxiv; 2026. Withdrawn preprint. Preprint. doi:10.64898/2026.01.09.698573.
10. Kharouiche O, Markogiannakis A, Fei X, Chatzianastasis M, Vazirgiannis M. Cell2Text: Multimodal LLM for Generating Single-Cell Descriptions from RNA-Seq Data; 2025. arXiv:2509.24840.
11. Markogiannakis A. Cell2Text: Multimodal LLM for generating textual descriptions from single-cell RNA-Seq profiles; 2026. Thesis, National Technical University of Athens. Available from: `https://dspace.lib.ntua.gr/xmlui/bitstream/handle/123456789/64348/thesis_ArisMarkogiannakis.pdf?sequence=1`.
12. Schaefer M, Peneder P, Malzl D, Peycheva M, Burton J, Hakobyan A, et al. Multimodal learning of transcriptomes and text enables interactive single-cell RNA-seq data exploration with natural-language chats. bioRxiv; 2024. Preprint. doi:10.1101/2024.10.15.618501.
13. Schaefer M, Peneder P, Malzl D, Lombardo SD, Peycheva M, Burton J, et al. Multimodal learning enables chat-based exploration of single-cell data. Nature Biotechnology. 2025;44(10):1722–1732. doi:10.1038/s41587-025-02857-9.
14. Shen X, Li X. OmniNA: A foundation model for nucleotide sequences. bioRxiv; 2024. Preprint. doi:10.1101/2024.01.14.575543.
15. Shen X, Li JX, Yang M, Shi L, Chen K, Li X. A foundation model for nucleotide sequences. Nucleic Acids Research. 2026;54(6). doi:10.1093/nar/gkag083.
16. Auer C, Lysak M, Nassar A, Dolfi M, Livathinos N, Vagenas P, et al. Docling Technical Report; 2024. arXiv:2408.09869.
17. Huang X, Qin J, Tang F, Yang C, Xu L, Wei J, et al. XunZi, an AI biologist, reveals disease-modifying targets. Nature Biomedical Engineering. 2026. doi:10.1038/s41551-026-01769-6.
18. Sun Q, Guo Z, Peng R, Chen H, Wang J. Any-to-Any Learning in Computational Pathology via Triplet Multimodal Pretraining; 2025. arXiv:2505.12711.
19. Gupta A, Buendia A, Kundaje A, Leskovec J. Bio-BLIP: A Multimodal Architecture for Transferable Reasoning in Genomic Variant Interpretation. bioRxiv; 2026. Preprint. doi:10.64898/2026.05.12.724740.
20. Al-Kateb G, Cengiz E, Gök M. BioGPT: A Generative Transformer-Based Framework for Personalized Genomic Medicine and Rare Disease Diagnosis. Mesopotamian Journal of Artificial Intelligence in Healthcare. 2025;2025:154–164. doi:10.58496/mjaih/2025/015.
21. Luo Y, Zhang J, Fan S, Yang K, Wu Y, Qiao M, et al. BioMedGPT: Open Multimodal Generative Pre-trained Transformer for BioMedicine; 2023. arXiv:2308.09442.
22. Fallahpour A, Magnuson A, Gupta P, Ma S, Naimer J, Shah A, et al. BioReason: Incentivizing Multimodal Biological Reasoning within a DNA-LLM Model; 2025. arXiv:2505.23579.
23. Tsou CH, Ozery-Flato M, Barkan E, Mahajan D, Shapira B. BioVERSE: Representation Alignment of Biomedical Modalities to LLMs for Multi-Modal Reasoning; 2025. arXiv:2510.01428.
24. Rizvi SA, Levine D, Patel A, Zhang S, Wang E, Perry CJ, et al. Scaling Large Language Models for Next-Generation Single-Cell Analysis. bioRxiv; 2025. Preprint. doi:10.1101/2025.04.14.648850.
25. He S, Levine D, Vrkic I, Bressana MF, Zhang D, Rizvi SA, et al. CaLMFlow: Volterra Flow Matching using Causal Language Models; 2024. arXiv:2410.05292.
26. Galkin F, Naumov V, Pushkov S, Sidorenko D, Urban A, Zagirova D, et al. Precious3GPT: Multimodal Multi-Species Multi-Omics Multi-Tissue Transformer for Aging Research and Drug Discovery. bioRxiv; 2024. Preprint. doi:10.1101/2024.07.25.605062.
27. Fang Y, Jin Q, Xiong G, Jin B, Zhong X, Ouyang S, et al. Cell-o1: Training LLMs to Solve Single-Cell Reasoning Puzzles with Reinforcement Learning; 2025. arXiv:2506.02911.
28. Gao Y, Wang W, Zhao Y, Dong K, Shan C, Zheng W, et al. Language may be all omics needs: Harmonizing multimodal data for omics understanding with CellHermes. bioRxiv; 2025. Preprint. doi:10.1101/2025.11.07.687322.
29. Fang Y, Liu K, Zhang N, Deng X, Yang P, Chen Z, et al. ChatCell: Facilitating Single-Cell Analysis with Natural Language; 2024. arXiv:2402.08303.
30. Ingraham JB, Baranov M, Costello Z, Barber KW, Wang W, Ismail A, et al. Illuminating protein space with a programmable generative model. Nature. 2023;623(7989):1070–1078. doi:10.1038/s41586-023-06728-8.

Proof

31. Jiang Q, Ye X, Guo Z, Xia Y, Liu Z, Xu J, et al. Phenotype-Guided In Silico Molecular Generation Using Large Language Models. bioRxiv; 2026. Preprint. doi:10.64898/2026.01.03.697483.
32. Tang Z, Wang F, He H, Zhou J, Lv T, Zhu J, et al. Aligning LLMs with Biomedical Knowledge using Balanced Fine-Tuning; 2025. arXiv:2511.21075.
33. Liu T, Xiao Y, Luo X, Xu H, Zheng WJ, Zhao H. Geneverse: A collection of Open-source Multimodal Large Language Models for Genomic and Proteomic Research; 2024. arXiv:2406.15534.
34. Liang W. Human Genome Book:Words,Sentences and Paragraphs. bioRxiv; 2025. Preprint. doi:10.1101/2025.01.23.634629.
35. Shen Y, Cao G, Wu J, Chen D, Feng C, Chen M. GenNA: Conditional generation of nucleotide sequences guided by natural-language annotations. bioRxiv; 2026. Preprint. doi:10.64898/2026.04.22.720063.
36. Danner M, Islam T, Begemann M, Kraft F, Elbracht M, Kurth I, et al. Genolator: A Multimodal Large Language Model Fusing Natural Language, Genomic, and Structural Tokens for Protein Function Interpretation. bioRxiv; 2025. Preprint. doi:10.1101/2025.11.14.688396.
37. Lyu Y, Wu Z, Zhang L, Zhang J, Li Y, Ruan W, et al. GP-GPT: Large Language Model for Gene-Phenotype Mapping; 2024. arXiv:2409.09825.
38. Levine D, Rizvi SA, Lévy S, Pallikkavaliyaveetil N, Zhang D, Chen X, et al. Cell2Sentence: Teaching Large Language Models the Language of Biology. bioRxiv; 2023. Preprint. doi:10.1101/2023.09.11.557287.
39. Gu Y, Wu Z, Yan R, Wang Z, Li Y, Lin S, et al. H2O: A Foundation Model Bridging Histopathology to Spatial Multi-Omics Profiling. bioRxiv; 2026. Preprint. doi:10.64898/2026.04.21.717342.
40. Liu J, Zhang P, Xu S, Srinivasan SS, Wu Y, Zhang J. Multimodal Cell Context Instruction Tuning for Conditional DNA Regulatory Sequence Generation with Large Language Models. In: 2025 IEEE International Conference on Image Processing (ICIP). IEEE; 2025. p. 989–994. doi:10.1109/icip55913.2025.11084625.
41. Shi Y, Yang J, Nai C, Li S, Fang J, Wang X, et al. Language-Enhanced Representation Learning for Single-Cell Transcriptomics; 2025. arXiv:2503.09427.
42. Zhavoronkov A, Aladinskiy V, Aliper A, Miftakhutdinov Z, Reymond M, Naumov V, et al. The End of Aging Clocks: Training Foundation Models to Reason in Aging and Longevity. bioRxiv; 2026. Preprint. doi:10.64898/2026.03.28.714980.
43. Tu T, Azizi S, Driess D, Schaekermann M, Amin M, Chang PC, et al. Towards Generalist Biomedical AI; 2023. arXiv:2307.14334.
44. Liu T, Xuan W, Wu H, Humphrey P, DiStasio M, Kahila M, et al. TeamPath: Building MultiModal Pathology Experts with Reasoning AI Copilots; 2025. arXiv:2511.17652.
45. Zhang C, Sun J, Xu Z, Liao R, Yin A, Gao H, et al. OCellus: A Language-Model Framework for Single-Cell, Spatial, and Perturbation Biology with Natural-Language Reasoning. bioRxiv; 2026. Preprint. doi:10.64898/2026.07.08.737248.
46. Sypetkowski M, Krawczyk J, Łukasz Smoliński, Kinas R, Pietrzak P, Jetka T, et al. OmicsLM: A Multimodal Large Language Model for Multi-Sample Omics Reasoning; 2026. arXiv:2605.06728.
47. Li Z, Subasri V, Shen Y, Li D, Zhao Y, Stan GB, et al. Omni-DNA: A Unified Genomic Foundation Model for Cross-Modal and Multi-Task Learning; 2025. arXiv:2502.03499.
48. Wang L. OmniGene-4: A Unified Bio-Language MoE Model with Router-Level Interpretability. bioRxiv; 2026. Preprint. doi:10.64898/2026.05.12.724542.
49. Xiang H, Ma P, Cao Y, Yu D, Chen H, Yang X, et al. Rethinking Genomic Modeling Through Optical Character Recognition; 2026. arXiv:2602.02014.
50. Queen O, Huang Y, Calef R, Giunchiglia V, Chen T, Dasoulas G, et al. ProCyon: A multimodal foundation model for protein phenotypes. bioRxiv; 2024. Preprint. doi:10.1101/2024.12.10.627665.
51. Fesser L, Zhang H, Li MM, Wang E, Perozzi B, Azizi S, et al. How Post-Training Shapes Biological Reasoning Models; 2026. arXiv:2606.16517.
52. Li G, You Y, Fu Y, Zhou W, Tang F, Kong J, et al. RVQ-Alpha: Bridging Single-Cell Transcriptomics and Large Language Models via Hierarchical Discrete Tokenization and Fact-Aware Reinforcement Learning. bioRxiv; 2026. Preprint. doi:10.64898/2026.04.20.719773.
53. Tang W, Liu R, Wen H, Dai X, Ding J, Li H, et al. A General Single-Cell Analysis Framework via Conditional Diffusion Generative Models. bioRxiv; 2023. Preprint. doi:10.1101/2023.10.13.562243.
54. Xiao X, Li Y, Zeng Z, Yan Y, Liu Z, Liu Z, et al. SciCore-Omics: a tri-modal foundation model unifying histology, spatial transcriptomics and language for spatial biology. bioRxiv; 2026. Preprint. doi:10.64898/2026.05.30.728937.
55. Shang Y, Tan J, Zeng X, Zhou R. scMIR: a vision-language foundation model for single-cell light microscopy image representation; 2026. arXiv:2607.22712.
56. Wei R, Zhang Z, Sun J, Sun Y, Meng J, Zheng P, et al. scMOBA: A conversational single-cell Multi-Omics Brain Agent across species. bioRxiv; 2025. Preprint. doi:10.64898/2025.12.01.691565.
57. Zhang T, Yu J, Lou S, Liang Z, Liang Y, Li Z, et al. Systematic discovery of single-cell protein networks in cancer with Shusi. bioRxiv; 2025. Preprint. doi:10.1101/2025.04.27.649905.
58. Liu S, Tang J, Ma J, Liang S. TissueNarrator: Generative Modeling of Spatial Transcriptomics with Large Language Models. bioRxiv; 2025. Preprint. doi:10.1101/2025.11.24.690325.
59. Chen W, Ye M, Xu T, Huang D, Zhang H, Li H, et al. CellTosg2Sequence: A Unified Text-Omics-Signaling-Graph Large Language Model for Single-Cell Analysis. bioRxiv; 2026. Preprint. doi:10.64898/2026.06.16.732397.
60. Pang M, Roy TK, Wu X, Tan K. Query-driven generative AI synthesizes multi-modal spatial omics from histology. bioRxiv; 2025. Preprint. doi:10.64898/2025.12.11.693669.
61. Hu EJ, Shen Y, Wallis P, Allen-Zhu Z, Li Y, Wang S, et al. LoRA: Low-Rank Adaptation of Large Language Models; 2021. arXiv:2106.09685.
62. OpenAI, :, Hurst A, Lerer A, Goucher AP, Perelman A, et al. GPT-4o System Card; 2024. arXiv:2410.21276.
63. Touvron H, Martin L, Stone K, Albert P, Almahairi A, Babaei Y, et al. Llama 2: Open Foundation and Fine-Tuned Chat Models; 2023. arXiv:2307.09288.

64. Cui H, Wang C, Maan H, Pang K, Luo F, Duan N, et al. scGPT: toward building a foundation model for single-cell multi-omics using generative AI. Nature Methods. 2024;21(8):1470–1480. doi:10.1038/s41592-024-02201-0.
65. Theodoris CV, Xiao L, Chopra A, Chaffin MD, Al Sayed ZR, Hill MC, et al. Transfer learning enables predictions in network biology. Nature. 2023;618(7965):616–624. doi:10.1038/s41586-023-06139-9.
66. Lin Z, Akin H, Rao R, Hie B, Zhu Z, Lu W, et al. Evolutionary-scale prediction of atomic-level protein structure with a language model. Science. 2023;379(6637):1123–1130. doi:10.1126/science.ade2574.
67. Schaar AC, Tejada-Lapuerta A, Palla G, Gutgesell R, Halle L, Minaeva M, et al. Nicheformer: a foundation model for single-cell and spatial omics. bioRxiv; 2024. Preprint. doi:10.1101/2024.04.15.589472.
68. Dip SA, Zafor A, Paul BK, Shuvo UA, Emon MI, Wang X, et al. LLM4Cell: A Survey of Large Language and Agentic Models for Single-Cell Biology; 2025. arXiv:2510.07793.
69. Liu S, Li Y, Li Z, Gitter A, Zhu Y, Lu J, et al. A Text-guided Protein Design Framework; 2023. arXiv:2302.04611.
70. Pei Q, Zhang W, Zhu J, Wu K, Gao K, Wu L, et al. BioT5: Enriching Cross-modal Integration in Biology with Chemical Knowledge and Natural Language Associations; 2023. arXiv:2310.07276.
71. Pei Q, Wu L, Gao K, Liang X, Fang Y, Zhu J, et al. BioT5+: Towards Generalized Biological Understanding with IUPAC Integration and Multi-task Tuning; 2024. arXiv:2402.17810.
72. Xia Y, Jin P, Xie S, He L, Cao C, Luo R, et al. Nature Language Model: Deciphering the Language of Nature for Scientific Discovery; 2025. arXiv:2502.07527.
73. Zhou X, Han C, Zhang Y, Du H, Tian J, Su J, et al. Decoding the Molecular Language of Proteins with Evolla. bioRxiv; 2025. Preprint. doi:10.1101/2025.01.05.630192.
74. Taylor R, Kardas M, Cucurull G, Scialom T, Hartshorn A, Saravia E, et al. Galactica: A Large Language Model for Science; 2022. arXiv:2211.09085.
75. Wang Z, Zhang Q, Ding K, Qin M, Zhuang X, Li X, et al. InstructProtein: Aligning Human and Protein Language via Knowledge Instruction; 2023. arXiv:2310.03269.
76. Zhuo L, Chi Z, Xu M, Huang H, Zheng H, He C, et al. ProtLLM: An Interleaved Protein-Language LLM with Protein-as-Word Pre-Training; 2024. arXiv:2403.07920.
77. Abdine H, Chatzianastasis M, Bouyioukos C, Vazirgiannis M. Prot2Text: Multimodal Protein's Function Generation with GNNs and Transformers; 2023. arXiv:2307.14367.
78. Liu Z, Zhang A, Fei H, Zhang E, Wang X, Kawaguchi K, et al. ProtT3: Protein-to-Text Generation for Text-based Protein Understanding; 2024. arXiv:2405.12564.
79. Lv L, Lin Z, Li H, Liu Y, Cui J, Chen CYC, et al. ProLLaMA: A Protein Large Language Model for Multi-Task Protein Language Processing; 2024. arXiv:2402.16445.
80. Wang E, Schmidgall S, Jaeger PF, Zhang F, Pilgrim R, Matias Y, et al. TxGemma: Efficient and Agentic LLMs for Therapeutics; 2025. arXiv:2504.06196.
81. Zhang J, Ubas AA, de Borja R, Svensson V, Thomas N, Thakar N, et al. Tahoe-100M: A Giga-Scale Single-Cell Perturbation Atlas for Context-Dependent Gene Function and Cellular Modeling. bioRxiv; 2025. Preprint. doi:10.1101/2025.02.20.639398.

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[Co-author biographical notes.]

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